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DECENTRALIZED FEDERATED LEARNING OVER IMPERFECT CHANNELS

PROJECT REPORT

Submitted by

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Abstract

This project studies Decentralized Federated Learning (D-FL) over imperfect communication channels, focusing on how network topology, packet loss and communication asynchrony affect model convergence and communication efficiency. Conventional Federated Learning depends on a central coordinator and reliable communication, which becomes a limitation in distributed, resource-constrained networks; decentralized FL removes this central server and lets devices exchange model updates directly with their neighbours, but its performance now depends heavily on the underlying communication structure. The work is organised into three progressive methodologies. M1 implements gossip-based federated learning under link errors and asynchronous delay and evaluates its robustness across ring, grid and random topologies. M2 introduces an event-triggered asynchronous gossip protocol that transmits updates only when a node's model state changes beyond a threshold, reducing communication overhead while aiming to preserve accuracy. M3 performs a spectral-gap analysis of the mixing matrix associated with each topology to theoretically explain, and experimentally validate, why different network structures exhibit different convergence behaviour. All three methodologies are evaluated under a common experimental protocol on the MNIST dataset using a lightweight CNN, and compared using accuracy, convergence rate, communication cost, consensus error and packet-loss tolerance. The goal of this project is to build a reproducible, common framework for understanding how imperfect channels and network topology jointly govern the robustness and efficiency of decentralized federated learning.

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Introduction

Federated Learning (FL) allows multiple devices to train a shared model without directly sharing their raw data. In the conventional FedAvg approach, clients perform local training and send their models to a central server for aggregation; although local training reduces communication frequency, FL still depends on a central coordinator and reliable communication, which can become a limitation in distributed and resource-constrained networks. Decentralized Federated Learning (D-FL) removes the central server and allows devices to exchange model updates directly with neighbouring devices, which improves the distributed nature of the system and avoids dependence on a single coordinator. However, the learning process now depends on the communication structure between devices, making network topology and connectivity important factors in model convergence.

In practical wireless or distributed networks, communication links may experience packet loss, delays, asynchrony and unreliable connectivity. When model updates are lost or received at different times, devices may hold inconsistent model states, which can reduce aggregation quality and slow down or destabilize convergence. Studying D-FL under such imperfect communication channels is therefore important for understanding its practical robustness. This project builds a common, reproducible framework that progressively studies gossip-based D-FL under link errors, a communication-efficient event-triggered variant, and a spectral-gap analysis that explains how network topology governs convergence.

Project Scope

The Decentralized Federated Learning Over Imperfect Channels project aims at the automation and evaluation of the following:

a. Analysis of the impact of packet loss and communication asynchrony on decentralized federated learning convergence and accuracy.

b. Development of an event-triggered gossip mechanism that reduces unnecessary communication while maintaining model accuracy.

c. Comparison of ring, grid and random network topologies based on convergence speed, consensus error and communication cost.

d. Spectral-gap analysis of the mixing matrix associated with each topology to explain the relationship between network structure and convergence.

e. Evaluation of all proposed methods using a common protocol on accuracy, convergence rate, communication cost, model divergence and packet-loss robustness.

Key Contacts and Stakeholders

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Project Resource Requirements

1. Software Resource Requirements
2. Operating System: Windows 11, macOS, or Linux
3. Programming Language: Python 3.x
4. Deep Learning Framework: PyTorch
5. Numerical Computing: NumPy
6. Graph & Topology Library: NetworkX
7. Dataset: MNIST
8. Model: Lightweight Convolutional Neural Network (CNN)
9. Simulation Environment: Python-based gossip / packet-loss simulator
10. System Design Tools: Draw.io and Mermaid.js

1. Source-Derived Technical Background

Federated Learning baseline

Federated Learning (FL) allows multiple devices to train a shared model without directly sharing their raw data. In the conventional FedAvg approach, clients perform local training and send their models to a central server for aggregation. Although local training reduces communication frequency, FL still depends on a central coordinator and reliable communication, which can become a limitation in distributed and resource-constrained networks.

Why decentralized FL is relevant

Decentralized Federated Learning (D-FL) removes the central server and allows devices to exchange model updates directly with neighboring devices. This improves the distributed nature of the system and avoids dependence on a single coordinator. However, the learning process now depends on the communication structure between devices, making network topology and connectivity important factors in model convergence.

Why imperfect channels are difficult

In practical wireless or distributed networks, communication links may experience packet loss, delays, asynchrony, and unreliable connectivity. When model updates are lost or received at different times, the devices may have inconsistent model states. This can reduce aggregation quality and slow down or destabilize convergence. Therefore, studying D-FL under imperfect communication channels is important for understanding its practical robustness.

Why topology matters

In decentralized FL, devices communicate only with their neighboring nodes, so the network topology determines how quickly model information spreads through the system. Different topologies such as ring, grid, and random graphs provide different levels of connectivity and communication overhead. Therefore, comparing topologies helps determine how network structure affects convergence speed, accuracy, and communication efficiency.

Why gossip-based aggregation is relevant

Gossip-based FL allows neighboring devices to exchange and average their model updates without requiring a central server. This provides a simple mechanism for decentralized consensus and serves as a suitable baseline for evaluating different network conditions. In this project, gossip-based FL is first evaluated under link errors and different topologies before introducing communication-efficient improvements.

Why communication-efficient gossip matters

Frequent communication can improve model synchronization but increases communication overhead. To address this trade-off, event-triggered asynchronous gossip can allow devices to communicate only when their model changes sufficiently. This can reduce unnecessary transmissions while attempting to maintain learning accuracy and convergence performance.

Why spectral-gap analysis matters

The convergence behavior of decentralized learning is closely related to the mixing matrix associated with the network topology. The spectral gap provides an indication of how effectively information propagates through the network. Therefore, this project uses spectral-gap analysis to theoretically explain and experimentally validate why different topologies exhibit different convergence behaviors.

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2. Literature Survey: 25+ Relevant Papers

ID	Author / Year	Paper	Main Contribution	Relevance
P27	Sun, Li & Wang (2023)	Decentralized Federated Averaging	Proposes DFedAvg and studies convergence in decentralized topologies.	Provides the baseline decentralized aggregation algorithm for M1.
P26	Lalitha et al. (2019)	Peer-to-Peer Federated Learning on Graphs	Introduces P2P federated learning using graph-based neighbor communication.	Provides the foundation for your P2P topology-based D-FL architecture.
P22	Liu, Chen & Zhang (2022)	Decentralized Federated Learning: Balancing Communication and Computing Costs.	Studies the trade-off between local computation and communication rounds.	Supports your choice of local update/aggregation frequency and communication analysis.
P28	Gao, Thai & Wu (2023)	When Decentralized Optimization Meets Federated Learning.	Connects D-FL with consensus, mixing matrices and decentralized optimization theory.	Supports M3 spectral-gap and mixing-matrix analysis.
P24	Yuan et al. (2024)	Decentralized Federated Learning: A Survey and Perspective	Reviews recent D-FL methods and identifies future research directions.	Helps validate and position your research gap.
P23	Martínez Beltrán et al. (2023)	Decentralized Federated Learning: Fundamentals, State of the Art, Frameworks, Trends, and Challenges.	Provides a comprehensive overview of D-FL architectures, protocols and challenges.	Useful for background, terminology and literature survey.
P30	Chen et al. (2025)	Mobility-aware Decentralized FL with Joint Optimization of Local Iteration and Leader Selection for Vehicular Networks.	Studies D-FL with dynamic topology and mobility.	Useful as a future extension from fixed to dynamic topologies.
P25	Chung et al. (2025)	Decentralized Federated Learning with Non-IID Data: Challenges, Trends, and Future Opportunities.	Studies the effect of Non-IID data on decentralized FL convergence.	Useful for discussing limitations/future work involving Non-IID + channel errors.
P-29	Nguyen et al. (2023)	Self-Organizing Democratized Learning: Toward Large-Scale Distributed Learning Systems.	Proposes distributed learning architectures for large-scale decentralized systems.	Supports scalability discussion and future expansion.
P-11	Jiang, Y. (2022)	Decentralized Edge Learning via Unreliable Device-to-Device Communications	Proposes a decentralized edge learning approach where devices learn collaboratively without a central server.	Supports robust and reliable model training even when devices have unstable connections.
P-12	Chen,S.(2022)	Decentralized Wireless Federated Learning With Differential Privacy.	Proposes a decentralized federated learning framework for wireless networks.	Proposes a decentralized federated learning framework for wireless networks.
P-13	Tang (2023)	GossipFL: A Decentralized Federated Learning Framework With Sparsified and Adaptive Communication	Proposes GossipFL, a decentralized federated learning framework.	Useful for wireless and resource-constrained edge/IoT devices.
P-14	Hossain, E. (2024)	Decentralized Federated Learning Over Imperfect Communication Channels.	Proposes a decentralized federated learning approach for wireless networks with imperfect communication channels.	Useful for edge computing, IoT, and mobile networks where communication channels are unreliable.
P-15	Yang, H. (2020)	Scheduling Policies for Federated Learning in Wireless Networks	Optimizes device selection and communication resources during training.	Relevant to wireless federated learning and resource optimization.
P-16	Salehi, M.(2021)	Federated Learning in Unreliable and Resource-Constrained Cellular Wireless Networks	Develops methods to improve FL performance despite unstable wireless connections.	Useful for cellular, IoT, and mobile networks with limited resources.

P-17	Yang, Z (2021)	Learning Rate Optimization for Federated Learning Exploiting Over-the-Air Computation	Optimizes the learning rate according to wireless channel conditions.	Helps improve communication-efficient model training.
P-18	Amiri,M.M (2022)	Convergence of Federated Learning Over a Noisy Downlink	Analyzes how channel noise and communication errors affect model updates.	Useful for edge and cellular networks with unreliable downlink communication.
P-19	Tang,T. (2022)	Fairness-Aware Federated Learning With Unreliable Links in Resource-Constrained Internet of Things	Addresses unreliable communication links between IoT devices.	Useful for resource-constrained wireless/edge devices.
P-1	Kavalionak, H., Carlini, E., Dazzi, P., Ferrucci, L., Mordacchini, M., Coppola, M. (2021)	Impact of Network Topology on the Convergence of Decentralized Federated Learning Systems	Studies how different network topologies and communication parameters affect convergence in decentralized federated learning.	Directly relevant to topology comparison and convergence analysis in M3.
P-2	Dötzer, M., Mao, Y., Diepold, K. (2023)	Communication Topologies for Decentralized Federated Learning	Compares decentralized and hierarchical topologies and relates topology structure to accuracy and communication overhead.	Supports systematic topology and communication-efficiency analysis.
P-3	Kalra, S., Wen, J., Cresswell, J. C., Volkovs, M., Tizhoosh, H. R. (2023)	Decentralized Federated Learning Through Proxy Model Sharing	Introduces ProxyFL, where clients exchange proxy models to enable decentralized collaboration with reduced communication overhead and model heterogeneity.	Relevant to communication-efficient decentralized FL and privacy-aware model exchange.
P-4	Gauthier, F., Gogineni, V. C., Werner, S., Huang, Y. F., Kuh, A. (2023)	Asynchronous Online Federated Learning with Reduced Communication Requirements	Studies asynchronous federated learning with delayed updates and partial-sharing communication to reduce communication overhead.	Relevant to M2's asynchronous communication and communication-efficiency analysis.
P-5	Hegedűs, I., Danner, G., Jelasity, M. (2021)	Decentralized Learning Works: An Empirical Comparison of Gossip Learning and Federated Learning	Provides an empirical comparison of gossip learning and federated learning, including communication, compression and convergence behavior.	Supports the gossip-based baseline used in M1.
P-6	Palmieri, L., Boldrini, C., Valerio, L., Passarella, A., Conti, M. (2024)	Impact of Network Topology on the Performance of Decentralized Federated Learning	Investigates how different network structures affect knowledge spreading and performance in decentralized federated learning.	Directly supports systematic network-topology comparison.
P-7	Chen, L., Liu, W., Chen, Y., Wang, W. (2024)	Communication-Efficient Design for Quantized Decentralized Federated Learning	Develops communication-efficient decentralized FL using quantization and adaptive communication to reduce communicated information.	Relevant to reducing communication overhead and improving communication efficiency.
P-8	Masmoudi, N., Jaafar, W. (2025)	OCD-FL: A Novel Communication-Efficient Peer Selection-Based Decentralized Federated Learning	Proposes peer selection for decentralized FL to reduce communication and energy consumption while maintaining learning performance.	Relevant to communication reduction and resource-constrained decentralized networks.

3. Comparative Synthesis

The literature provides strong foundations in decentralized federated learning, peer-to-peer model aggregation, communication-efficient learning, and decentralized optimization. However, these aspects are often studied separately. The proposed project uses a staged comparison to study the effect of imperfect communication, reduce unnecessary communication, and explain the role of network topology under a common experimental setting.

Dimension	M1	M2	M3	Literature gap addressed
Core mechanism	Gossip-based FL + imperfect channels.	Event-triggered asynchronous gossip.	Mixing matrix + spectral analysis.	Need for a controlled three-stage D-FL comparison.
Communication	Packet loss + delay	Communicate only when model change > threshold	Analyze how topology affects information mixing.	Channel imperfections and communication efficiency are often studied separately.
Topology	Ring, Grid, Random	Same topologies for fair comparison	Ring, Grid, Random	Limited systematic topology comparison under imperfect channels.
Main objective	Measure effect of packet loss/asynchrony on convergence	Reduce communication while maintaining accuracy.	Explain convergence differences between topologies.	Connect robustness, communication
				efficiency, and topology.
Key metric	Accuracy, convergence rate, model divergence, packet-loss tolerance	Accuracy, messages, bytes, convergence	Spectral gap, convergence rate, accuracy	Common reproducible evaluation across all methods.
Baseline	Ideal channel vs imperfect channel	Normal gossip vs event-triggered gossip.	Observed convergence vs spectral-gap prediction.	Fair same-scenario evaluation.

4. Proposed Research Gap

Existing studies have mainly focused on decentralized federated learning, gossip-based communication, communication efficiency, and convergence analysis separately. There is limited work that studies these factors together under imperfect communication channels. In particular, the combined effect of packet loss, asynchrony, network topology, and communication overhead on decentralized FL is not sufficiently evaluated in a common experimental setting. Therefore, this project proposes a three-stage approach: M1 studies the effect of packet loss and asynchrony, M2 reduces unnecessary communication using event-triggered gossip, and M3 analyzes the effect of topology using mixing matrices and spectral-gap analysis.

5. Problem Statement

Decentralized federated learning removes the central server by allowing devices to exchange model updates with their neighbors. However, packet loss, communication delays, asynchrony, and different network topologies can affect model convergence and accuracy. Frequent communication also increases communication overhead. Therefore, the problem is to develop a common D-FL framework that studies the effect of imperfect channels, reduces unnecessary communication using event-triggered gossip, and analyzes how network topology affects convergence using spectral-gap analysis.

6. Research Objectives

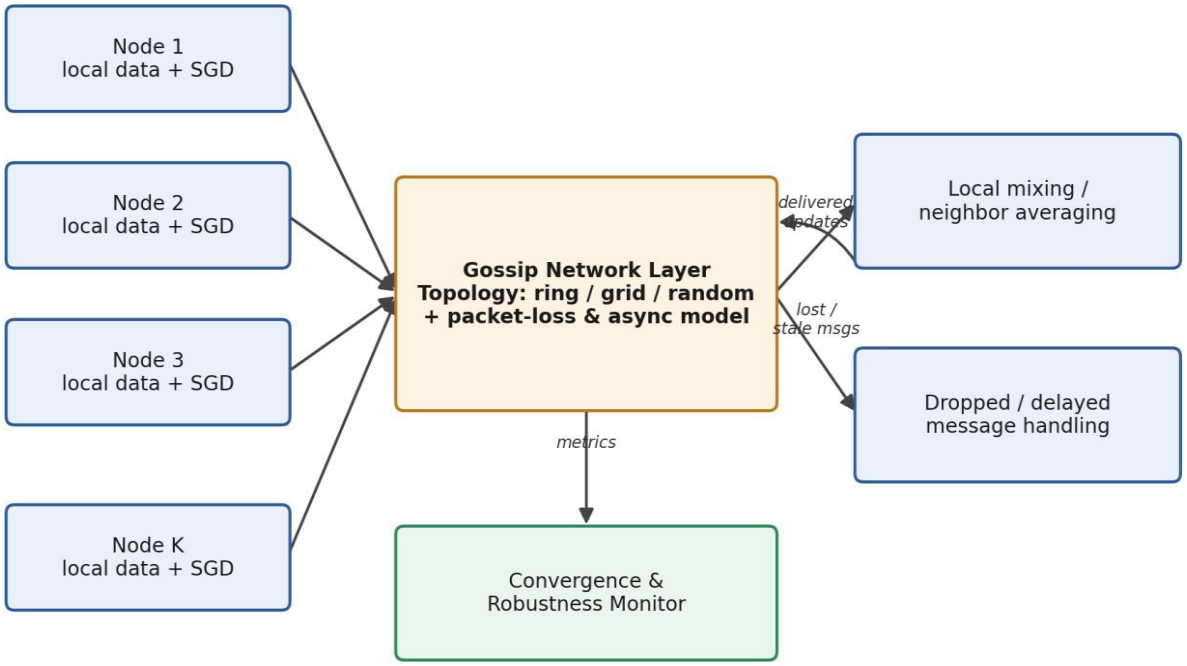
a. To analyze the impact of packet loss and communication asynchrony on decentralized federated learning.
b. To develop an event-triggered gossip mechanism to reduce unnecessary communication while maintaining model accuracy.
c. To compare different network topologies based on convergence and communication

performance.

d. To analyze the mixing matrix and spectral gap to understand the relationship between network topology and convergence.
e. To evaluate the proposed methods using accuracy, convergence rate, communication cost, model divergence, and packet-loss robustness.

7. Methodology Architecture

The following three architectures map directly to the proposed M1–M3 progression: M1 establishes the general robustness testbed under link errors and asynchrony; M2 builds an efficient communication protocol on top of it; M3 supplies the theoretical explanation (spectral gap) that connects topology choice to the robustness observed in M1 and the efficiency gains observed in M2.



M1: Gossip-Based FL under Link Errors and Asynchrony (Baseline Robustness)

Each node trains locally on its own data partition with SGD and exchanges model updates with its neighbors through a gossip network layer. The topology (ring, grid, random) and a packet-loss / asynchronous-delay model determine which updates are delivered, dropped, or delayed. Delivered updates are locally mixed (neighbor averaging); dropped or stale messages are logged and handled separately. A convergence and robustness monitor continuously records consensus error, divergence, and accuracy so that topologies and mixing schemes can be compared under identical error conditions.

M2: Event-Triggered Asynchronous Gossip Protocol

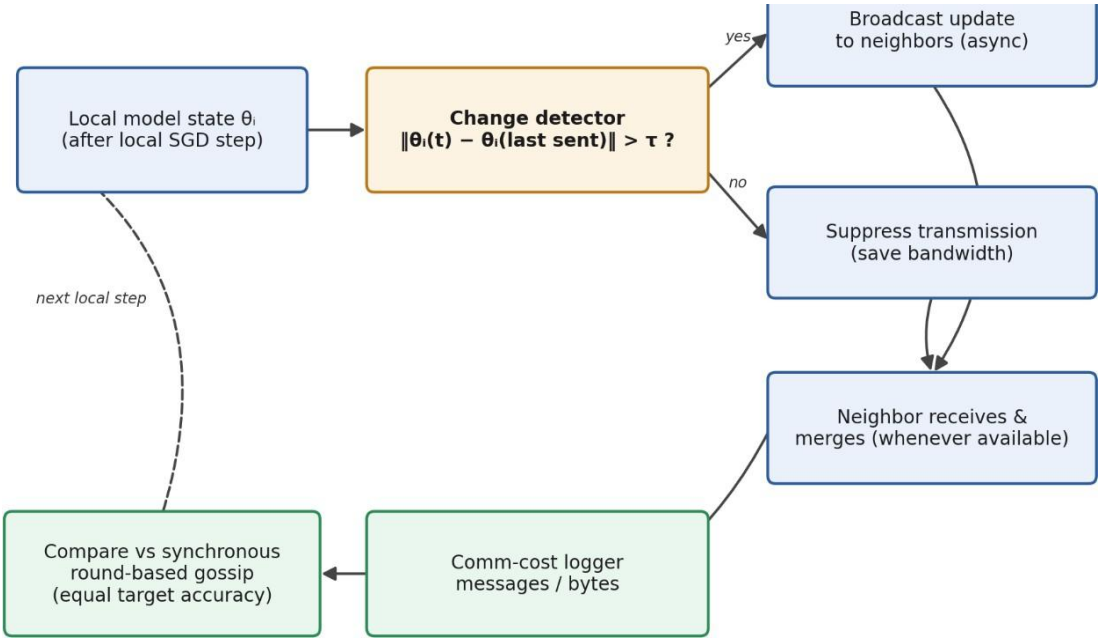


Figure 2. M2 architecture.

After each local SGD step, a node's model state is checked by a change detector: if the change since the last transmission exceeds threshold τ, the update is broadcast asynchronously to neighbors; otherwise transmission is suppressed to save bandwidth. Neighbors merge whichever updates they have received whenever they become available (no fixed round boundary). A communication-cost logger tracks messages and bytes exchanged, which are then compared against a synchronous round-based gossip baseline at the same target accuracy to quantify the communication savings of event-triggering.

M3: Spectral Analysis of the Mixing Matrix

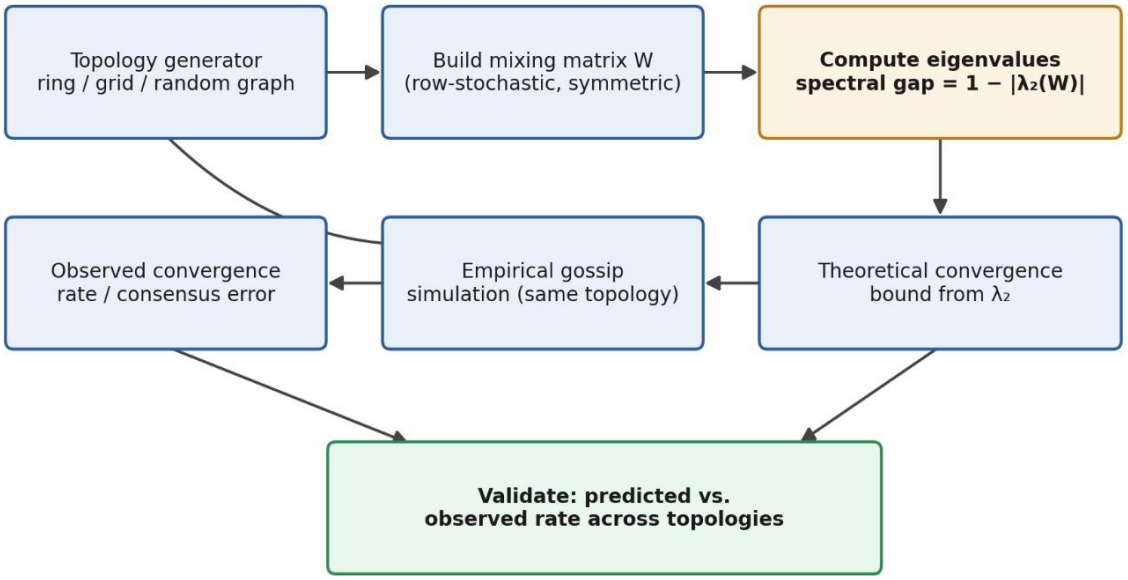


Figure 3. M3 architecture.

For each candidate topology (ring, grid, random graph), a row-stochastic mixing matrix W is constructed and its eigenvalues computed, yielding the spectral gap $1 - |\lambda_2(W)|$. This value feeds a theoretical convergence bound, which is compared against the convergence rate and consensus error observed empirically from a gossip simulation run on the same topology. The validation step checks whether the predicted ordering and rate of convergence across topologies matches what is actually observed, including under the packet-loss and asynchrony conditions introduced in M1.

8. Review 1 Experimental Plan

Item	Plan
Common setup	Python + PyTorch + NumPy; NetworkX for topology generation; fixed random seeds for reproducibility.
Dataset/model	MNIST dataset; lightweight CNN model; 5 virtual clients with local training.
Data regimes	IID and non-IID data partitions; identical data partitions used for M1–M3 comparison.
Network topology	Ring, Grid and Random peer-to-peer topologies with neighbor-based model exchange.

Channel	Imperfect links with configurable packet-loss rates and link failures.
M1 experiment	Evaluate decentralized Gossip FL under packet loss; compare topology-wise test accuracy, convergence speed, consensus error and communication cost.
M2 experiment	Event-triggered asynchronous gossip using model-change threshold; compare communication reduction, convergence and final accuracy with M1.
M3 experiment	Construct mixing matrix for each topology; calculate eigenvalues and SLEM; compare spectral predictions with experimental convergence rate.
Evaluation	Test accuracy, convergence rounds, consensus error, packet-loss tolerance, messages/bytes transmitted and communication savings.

9. Review 1 Deliverables Checklist

1. Literature survey covering at least 25 papers
2. Paper comparison matrix grouped by FL, AirComp, RIS, device selection, energy/latency and non-IID handling
3. Architecture diagrams for M1, M2 and M3
4. Research gap, problem statement and measurable objectives
5. Common experimental protocol and evaluation metrics
6. Preliminary baseline implementation plan and reproducibility checklist

10. Key Risks and How to Frame Them

Novelty risk: Similar Gossip FL and communication-efficient methods exist. Frame the work as a comparative study of decentralized FL under imperfect channels.

Scope risk: Complete M1 first, reuse it for M2, and implement M3 using mixing-

matrix/SLEM analysis. Simulation risk: Use a Python-based simulator for packet loss

instead of a full wireless simulator initially.

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11. Selected Source Mapping

- FedAvg/FL: FL basics, non-IID data and communication.
- Gossip/Decentralized FL: Peer-to-peer model exchange.

- Event-triggered FL: Communication reduction for M2.
- Imperfect-channel FL: Packet loss and unreliable links for M1.
- Spectral/Consensus studies: Mixing matrix and SLEM analysis for M3.

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12. Conclusion

The literature review establishes the importance of decentralized federated learning, communication efficiency, and reliable model exchange under imperfect network conditions. Based on the identified research gap, this project follows a progressive three-stage methodology. M1 investigates gossip-based federated learning under packet loss and communication asynchrony. M2 extends this approach through event-triggered asynchronous gossip to reduce unnecessary communication while maintaining learning performance. M3 analyses the mixing matrix and spectral gap to study the relationship between network topology and convergence. The proposed study evaluates these methodologies using common performance measures, including model accuracy, convergence rate, communication cost, consensus error, and packet-loss tolerance across different network topologies. This provides a structured foundation for the subsequent implementation and experimental evaluation of decentralized federated learning over imperfect communication channels.

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